

University of the Witwatersrand
IDVE Quiz - COMS4060A/COMS7056A
Date: September 19, 2025

Please read the following instructions carefully.

- You have 45 minutes to complete this quiz. You must attempt all question to score maximum marks.
- All the questions are multiple-choice and each question has only **ONE** correct answer.
- There is a total of 30 MCQs with **no negative marking**.
- Each questions counts for **2 marks**, giving a total of 60 marks available.
- It is in your best interest to **read the instructions for individual questions carefully** before answering.

Multiple Choice Questions

Question 1

What is the primary purpose of data visualization?

- A. To create visually appealing graphs
- B. To make sense of large data, identify patterns and capture insights**
- C. To evaluate data storage requirements
- D. To increase the dimensionality of data

Question 2

Which of the following statements are FALSE about descriptive statistics?

- A. They can effectively hide the shape of the data distribution
- B. They are primarily used for complex inferential analysis**
- C. They include measures like mean, standard deviation and variance
- D. They are useful for getting a quick overview of the data

Question 3

In a distribution with negative skewness, which of the following statements are TRUE?

- A. The tail on the left side is longer than the tail on the right side**
- B. The tail on the right side is longer than the tail on the left side
- C. The tails on either side are equal
- D. The mean is greater than the mode

Question 4

If the correlation coefficient between 2 random variables X and Y is -0.8, which of the following statements are TRUE?

- A. As X increases, Y increases with a strong relationship
- B. As X increases, Y decreases with a weak relationship
- C. As X increases, Y decreases with a strong relationship**
- D. X and Y have no relationship with each other

Question 5

What is the primary difference between a Kernel Density Estimation (KDE) and a traditional histogram?

- A. KDE uses only discrete data points, while histograms use continuous data
- B. KDE shows exact probabilities while histograms show density
- C. KDE represents each data point as a continuous curve and combines them for a smoothed estimation**
- D. KDE can only be used on Gaussian distributions

Question 6

Which of the following statements are TRUE about basic data preprocessing techniques?

- A. Standardisation scales data to have a mean of 1, and a variance of 0
- B. Normalisation scales features to have a fixed range**
- C. Normalisation transforms data to have 0 mean and unit variance
- D. None of the above

Question 7

What is the phenomenon called when removing outliers causes previously normal values to become new outliers?

- A. Masking**
- B. Variance reduction effect
- C. Nested outliers
- D. None of the above

Question 8

When dealing with categorical features in machine learning, which of the following statements is FALSE?

- A. Label encoding converts categorical values into a single column of numeric values
- B. Label encoding should always be used over one-hot encoding as it is more memory-efficient**
- C. One-hot encoding creates multiple new columns with binary values (0/1) for each category
- D. One-hot encoding avoids the assumption of ordinal relationships between categories

Question 9

Which of the following statistical measures is most resistant to the influence of outliers?

- A. Mean
- B. Standard Deviation
- C. Variance
- D. Median**

Question 10

Which outlier detection method is most appropriate when the data follows a Gaussian distribution, and you want to identify values based on their distance from the mean?

- A. Isolation forest**
- B. Histogram-based detection
- C. Z-score method
- D. Percentile-based removal

Question 11

When working with time-series data, which of the following statements is TRUE?

- A. There is always a correlation between adjacent data points
- B. Future values can always be perfectly predicted from past values
- C. The mean and variance change over time
- D. The order of observations contains important information**

Question 12

Which of the following operations is commonly applied to make a non-stationary series stationary?

- A. Exponentiation of the data
- B. Differencing successive observations**
- C. Randomly shuffling the data
- D. Adding a constant to every observation

Question 13

Why is stationarity an important assumption when building a machine learning model on time-series data?

- A. Because it guarantees that the series has no noise
- B. Because non-stationary data cannot be visualized in a meaningful way
- C. Because machine learning models rely on the data having a constant mean and variance**
- D. Because stationarity directly ensure seasonality is present

Question 14

Which of the following best describes seasonality in time-series data?

- A. A random fluctuation around the mean that has no pattern
- B. A repeating pattern in the data that occurs at fixed and predictable intervals**
- C. A long-term upward or downward movement in the mean level of the series
- D. A transformation applied to make the data stationary

Question 15

When testing stationarity, what conditions must be met to conclude that a process is stationary using the Augmented Dickey-Fuller (ADF) test?

- A. Only the test statistic must be lower than the critical value
- B. Only the p-value must be less than 0.05
- C. Either A OR B must be true
- D. Both A AND B must be true**

Question 16

When looking at time-series data, what does autocorrelation primarily measure?

- A. **The similarity between observations at different time lags**
- B. The absolute values of observations over time
- C. The correlation between time-dependent variables
- D. The seasonal temperature variations

Question 17

While selecting features using the Chi-Squared test, what is the correct interpretation when we "fail to reject" the null hypothesis?

- A. The feature is definitely independent of the target variable and should be kept
- B. The feature is definitely dependent on the target variable and should be kept
- C. **There is not enough evidence to suggest that the feature is useful, so we should discard it**
- D. We have enough evidence to indicate that the feature is useful, so we should keep it

Question 18

When dealing with categorical high-cardinality data, which of the following approaches are valid?

- A. One-hot encoding the categorical data
- B. Encoding the categories by assigning sequential integers (1, 2, 3,...)
- C. Replacing categories with continuous transform like supervised ratios or Weight of Evidence**
- D. Removing categories from the data until the cardinality becomes manageable

Question 19

Which of the following statements about time-series decomposition is TRUE?

- A. Seasonality refers to unpredictable and irregular fluctuations in the data
- B. Cyclical patterns occur at fixed and predictable intervals, such as every month or quarter
- C. Trend captures the long-term progression in the data, either upward or downward**
- D. Noise is a repeating component that occurs at regular intervals

Question 20

Which of the following statements about random walks is TRUE?

- A. A random walk is simply a series of unrelated random numbers
- B. A random walk is stationary because its mean and variance remain constant over time
- C. A random walk is generally non-stationary and shows strong autocorrelation at small time lags**
- D. Random walks are best modeled using seasonality and trend components

Question 21

Which of the following statements about preprocessing transformations is TRUE?

- A. Tree-based algorithms require features to be normally distributed before training
- B. The Box–Cox transformation can be applied to any dataset, including those with negative values
- C. Distance-based algorithms, such as nearest neighbours, are highly sensitive to feature scaling**
- D. Log transformations are mainly used to reduce seasonality in time-series data

Question 22

Which of the following indicates a non-stationary time-series?

- A. Autocorrelation decays quickly to zero
- B. The mean remains constant over time
- C. The variance changes over time or shows a trend**
- D. Data is normally distributed

Question 23

Which of the following statements about entropy are TRUE?

- A. A skewed probability distribution indicates high entropy
- B. A balanced probability distribution indicates low entropy
- C. A balanced probability distribution indicates high entropy**
- D. None of the above statements are true

Question 24

Which of the following statements correctly describes a type of feature selection method?

- A. Filter methods evaluate features independently of any machine learning model, usually using statistical measures like correlation or chi-square**
- B. Wrapper methods remove features randomly without considering model performance
- C. Embedded methods are applied after the model has been trained and do not affect training
- D. All feature selection methods always require deep learning models to function

Question 25

Which of the following statements correctly describes the Minimum-Redundancy-Maximum-Relevance (mRMR) feature selection method?

- A. It selects features randomly and then measures model performance
- B. It iteratively selects features that are highly relevant to the target while minimizing redundancy with already selected features**
- C. It selects features purely based on their correlation with each other
- D. It is similar to filter methods, evaluating each feature independently of others

Question 26

What is the key advantage of HDBSCAN over traditional clustering approaches like k-means?

- A. It requires fewer computational resources
- B. It requires a predefined number of clusters
- C. It works well with irregularly shaped clusters and noisy data**
- D. It performs better with clean, well-structured data

Question 27

Which of the following best describes spatial autocorrelation?

- A. It ignores the location of observations and only examines feature correlations
- B. It measures the degree to which a variable is similar or dissimilar to nearby locations**
- C. It is only applicable to time-series data
- D. It transforms categorical data into numerical features for machine learning

Question 28

Moran's I is a statistic used to measure global spatial autocorrelation. Which of the following statements is TRUE?

- A. A Moran's I value of 0 indicates strong positive spatial correlation
- B. A Moran's I value of 1 indicates perfect positive spatial correlation, 0 indicates no spatial correlation, and -1 indicates perfect negative correlation**
- C. Moran's I can only take positive values
- D. Moran's I does not consider the spatial arrangement of observations

Question 29

Which of the following statements about skewness and kurtosis are TRUE?

- A. Skewness measures the asymmetry of a distribution; a value of 0 indicates symmetry
- B. Kurtosis measures the heaviness of the tails of a distribution; higher kurtosis indicates more extreme values (outliers)
- C. A and B are true**
- D. None of the above

Question 30

Which of the following statements about Fourier Transforms in time series analysis is TRUE?

- A. A Fourier Transform converts a time series into a series of cumulative sums
- B. A Fourier Transform decomposes a time series into its constituent frequencies and their amplitudes, revealing periodic patterns such as seasonality**
- C. Fourier Transforms can only be applied to non-stationary data
- D. The frequency domain representation shows the data in units of time rather than cycles per unit time